OYM NBTS TEST-7 (Rolex)

1. In measurement of mass of a given object by the principle of moments, the meter scale is hung from its mid-point. A known weight of mass 4 kg is hung at one end of meter scale and a unknown weight of mass m kg is hung 20 cm from the centre at other side of the balance. The value of mass m is

a. 1.6 kg
b. 4 kg
c. 8 kg
d. 10 kg

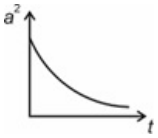
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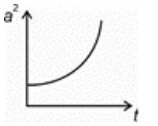
2. A pendulum with time period of 2 s is losing energy due to damping. At certain time its energy is 50 J. If after completing 10 oscillations, its energy has become 25 J, its damping constant (in kg s^{-1}) is (Given mass of bob = 1 kg)

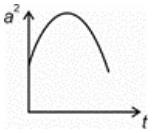
a. $\frac{1}{40}\ln 2$
b. $\frac{1}{20}\ln 2$
c. $\frac{1}{20}\ln 4$
d. $\frac{1}{40}\ln 8$

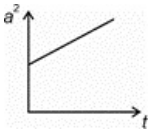
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3. To study the dissipation of energy of a simple pendulum, students plots a graph between square of amplitude and time. The correct graph is given by

a. 

b. 

c. 

d. 

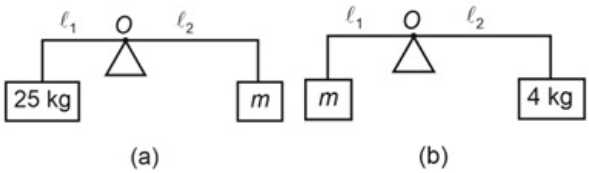
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4. In the damped oscillation with very small amount of damping, frequency of oscillation

a. Decreases continuously
b. Increases continuously
c. Almost remains constant
d. May decrease or increase

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5. In an experiment with a beam balance, an unknown mass m is balanced by two known masses of 25 kg and 4 kg as shown in figures (a) and (b)



The unknown mass m is

a. 10 kg
b. 15 kg
c. 18 kg
d. 20 kg

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6. In which direction does the damping force acts?

a. Opposite to the motion
b. Along the motion
c. Perpendicular to motion
d. Always in a fixed direction

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7. Vernier constant is given by (The symbols have their usual meaning)

a. MSD + VSD
b. VSD + 2 MSD
c. MSD – VSD
d. $\frac{\text{MSD}}{\text{VSD}}$

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8. In an experiment the angles are required to be measured using an instrument in which 59 divisions of the main scale coincide with the 60 division of vernier scale. If the smallest division of the main scale is 1 degree, then least count of the instrument is

a. One minute
b. Half minute
c. One degree
d. Half degree

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9. There are 20 division on the main scale in 1 cm and circular scale has 50 division. The least count of the scale is

a. 0.01 mm
b. 0.001 mm
c. 0.1 mm
d. 0.02 mm

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10. A micrometre screw has a negative zero error of 4 division. While measuring the diameter of a wire, the reading on the main scale is 3 division and 25th circular scale division coincide with base line. If the least count is 0.01 mm and number of division on the main scale are 10 in a cm, then correct diameter is

a. 0.339 cm
b. 0.329 cm
c. 0.319 cm
d. 0.320 cm

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11. If the Young's modulus of a metal wire of cross-section area 4 mm^2, experiencing 10^5 N of force is 10^{11} N/m^2, then strain produced in wire is

a. 0.05
b. 0.50
c. 0.25
d. 2.5

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12. Young's modulus of a material is defined as

a. $\frac{\text{Shear stress}}{\text{Shear strain}}$
b. $\frac{\text{Normal stress}}{\text{Longifudinal strain}}$
c. $\frac{\text{Normal force}}{\text{Volume}}$
d. $\frac{\text{Normal stress}}{\text{Shear stress}}$

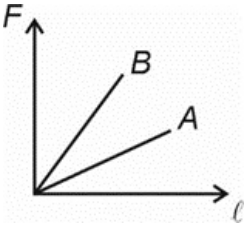
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13. The following four wires are made of the same material. Which one of these will have largest extension when the same tension is applied?

a. Length = 50 cm, diameter = 0.5 mm
b. Length = 100 cm, diameter = 1 mm
c. Length = 150 cm, diameter = 1.5 mm
d. Length = 200 cm, diameter = 2 mm

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14. Two wires A and B of same length are made of same material. In a experiment to determine Young's modulus graph between load F and extension ℓ of two wires are drawn. Then



a. The cross-sectional area of A is greater than that of B
b. The elasticity of B is greater than that of A
c. The cross-sectional area of B is greater than that of A
d. Young's modulus of A is greater than that of B

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15. The factors affecting the surface tension of a liquid is/are

a. Nature of liquid
b. Surrounding fluid
c. Temperature of liquid
d. All of these

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16. A liquid rises to height h in a certain tube with R as the radius of concave meniscus. Now the tube length is reduced below h, then liquid in tube

a. Will over flow
b. Will rise to maximum length with R as the radius of concave meniscus
c. Will rise to maximum length with concave meniscus of radius $R < R$
d. Will rise to maximum length with concave meniscus of radius $R > R$

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17. The property of liquid by virtue of which it opposes the flow of itself is called

a. Bulk modulus
b. Shear modulus
c. Surface tension
d. Viscosity

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18. A small steel ball of radius r is allowed to fall under gravity through a column of viscous liquid of coefficient of viscosity η. After some time the velocity of the ball attains a constant speed called terminal velocity. The coefficient of viscosity depends upon (i) (m) mass of the ball (ii) v_T (iii) g (iv) r (radius of ball).

Which of the following relations is dimensionally correct?

a. $\eta \propto \frac{mg}{(v_T)r}$
b. $\eta \propto \frac{(v_T)r}{mg}$
c. $\eta \propto \frac{rmg}{(v_T)}$
d. $\eta \propto mgr$

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19. The terminal speed of a copper ball of radius 2 mm falling through a tank of oil is 6.5 cm/s. If the density of oil is 2.4×10^3 kg/m^3 and that of copper is 8.9×10^3 kg/m^3, then coefficient of viscosity is nearly [take $g = 10$ ms^{-2}]

a. 0.9
b. 0.7
c. 1.8
d. 1.4

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20. In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first resonance is observed when the air column has a length of 20.0 cm for a tunning fork of frequency 400 Hz is used. The velocity of sound in air is 336 ms^{-1}. The second resonance is observed when the length of air column is

a. 60 cm
b. 62 cm
c. 63 cm
d. 65 cm

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21. In the experiment to determine the speed of sound using the resonance column,

a. The prongs of the tuning fork are kept in vertical plane
b. The prongs of the tuning fork are kept in horizontal plane
c. At first resonance, the length of the resonating

22. To determine the specific heat capacity of a given solid and liquid by the method of mixture following precaution should be taken. Choose the correct precautions among the following.

a. Physical balance should be in proper working condition
b. Take small amount of liquid such that solid does not get submerged in it

air column is close to the one-forth of the wavelength
d. Both (1) & (3)

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c. The solid used should not be chemically reactive
d. Both (1) and (3)

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23. 100 g of water at 40°C is added to a calorimeter containing 50 g of water at 20°C. The final temperature at equilibrium is 30°C. Calculate the heat capacity of calorimeter. Given specific heat capacity of water is 4.2 J g^{-1}°C^{-1}.

a. 105 J/°C
b. 840 J/°C
c. 210 J/°C
d. 110 J/°C

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24. Two different liquids at different temperatures are mixed. The final temperature of the mixture

a. Can be more than temperature of hotter liquid
b. Can be less than temperature of colder liquid
c. Can be more than temperature of colder liquid
d. Can never be less than temperature of hotter liquid

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25. In a metre bridge experiment, it is recommended to have known and unknown resistance of the similar values. This recommendation

a. Reduces error in finding the null point
b. Makes null point to lie towards the right of centre of wire
c. Reduce sensitivity of the galvanometer
d. Reduces current through galvanometer

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26. In a meter bridge experiment, the known and unknown resistance are interchanged to remove

a. Index error
b. End correction error
c. Error due to temperature effect
d. Random error

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27. In a experiment to verify Ohm's law, a student measured the potential drop across the resistor as 3.90 V and current through it 1.3 A. The most suitable way to report the resistance is

a. 3.0 Ω
b. 3.00 Ω
c. 2.90 Ω
d. 3.1 Ω

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28. A diminished image of a real object is to be obtained on a screen 1.0 m from it. This can be achieved by

a. A concave mirror of suitable focal length
b. A convex mirror of suitable focal length
c. A convex lens of suitable focal length
d. Both (1) & (3)

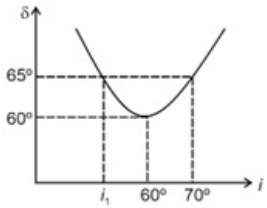
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29. Choose the incorrect statement among the following.

a. The mirror equation establish relation between u, v and f
b. Real and inverted image with same size is obtained if the object is placed on the distance $2f$ from the pole of a concave mirror
c. The image formed by concave mirror is always real
d. Convex mirror have larger field of view than concave mirror

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30. The angle of deviation (δ) versus angle of incidence (i) is plotted for a prism



The angle of prism is equal to

a. 60°
b. 75°
c. 90°
d. 120°

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31. The voltage at which forward bias current increases rapidly is called

a. Break down voltage
b. Forward voltage
c. Knee voltage
d. Voltage barrier

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32. The leakage current in forward bias Pn junction diode is measured generally in

a. A
b. mA
c. μA
d. nA

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33. In a circuit the load current is 20 mA and the unregulated voltage is 20 V. If a Zener diode of 5 V is connected parallel to the load, then current through load resistance becomes

a. 15 mA
b. 10 mA
c. 7.5 mA
d. 5 mA

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34. In an experiment of half deflection method if $R = 18\ \Omega$ and $S = 12\ \Omega$ for half deflection, then galvanometer resistance (G) is equal to

a. 18 Ω
b. 36 Ω
c. 27 Ω
d. 54 Ω

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35. Figure of merit of a galvanometer is 0.1 mA per division. If the galvanometer has 10 divisions on either side of zero the full deflection current in one side is

a. 2 mA
b. 20 mA
c. 1 mA
d. 10 mA

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36. The bridge wire in meter bridge is made of material like constantan or maganin because

a. They have small resistivity and small temperature coefficient of resistivity
b. They have large resistivity and large temperature coefficient of resistivity
c. They have large resistivity and small temperature coefficient of resistivity
d. They have small resistivity and large temperature coefficient of resistivity

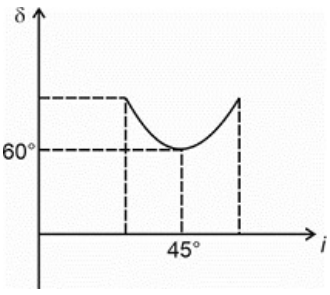
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37. A screen is placed at a distance 40 cm away from an illuminated object. A converging lens is placed between the source and screen and it is attempted to form the image of the source on the screen. If no position could be found, then focal length of the lens

a. Must be less than 10 cm
b. Must be greater than 20 cm
c. Must not be greater than 20 cm
d. Must be greater than 10 cm

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38. The curve of angle of deviation versus angle of incidence has been plotted for a prism. The value of refractive index of the prism used is



a. $\sqrt{3}$
b. $\frac{2}{\sqrt{3}-1}$
c. $\frac{\sqrt{3}}{2}$
d. $\frac{2}{\sqrt{3}+1}$

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39. In a resonance experiment to determine the speed of sound in air, a pipe of diameter 5 cm is used. The column in pipe resonates with tuning fork of frequency 480 Hz, when the minimum length of air column is 16 cm. The next resonance occur at ℓ_2 = 51 cm. The speed of sound wave is

a. 336 m s^{-1}
b. 340 m s^{-1}
c. 344 m s^{-1}
d. 330 m s^{-1}

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40. While determining the surface tension of water by capillary rise some detergent is added to pure water. Then rise of detergent water in capillary w.r.t pure water

a. Will increase
b. Will decrease
c. Will remain same
d. May increase or remain same

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41. In a Searle's experiment, the diameter of wire as measured by screw gauge of least count 0.001 cm is 0.050 cm. The length, measured by a scale of least count of 0.1 cm, is 100.0 cm. When weight of 100 N is suspended from the wire, the extension measured is 0.100 cm by a micrometre of least count 0.001 cm. The percentage relative error in the measurement of Young's modulus of material of wire is

a. 4.2%
b. 5.5%
c. 2.2%
d. 5.1%

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42. Moment of force depends upon

a. Magnitude of force
b. Perpendicular distance from pivot
c. Axis of rotation
d. All of these

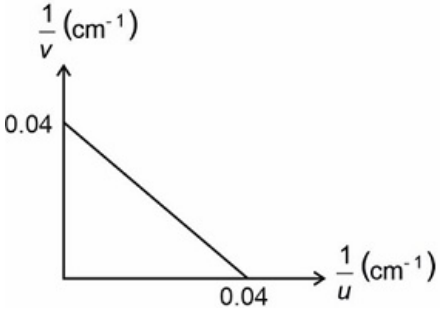
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43. In a physical balance working on the principle of moments, when 5 mg weight is placed in left pan, the beam becomes horizontal. Both the empty pans of the balance are of equal mass. Which of the following statement is correct?

a. Left arm is longer than the right arm
b. Both the arm are of same length
c. Left arm is shorter than the right arm
d. Every object that is weighed using this balance appears lighter than its actual weight

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44. The focal length (f) of a thin convex lens is determine by u – v method. The numerical values of the image distance (v) are measured corresponding to various object distances (u). The graph of $\frac{1}{v}$ (in cm^{-1}) versus $\frac{1}{u}$ (in cm^{-1}) is shown below. The power of lens is



a. 2 D
b. 4 D
c. 8 D
d. 10 D

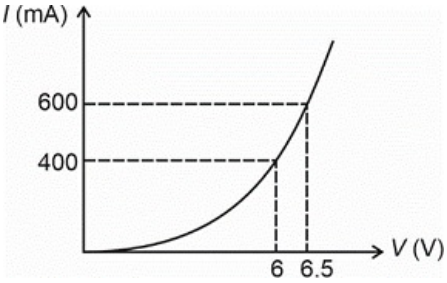
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45. Figure of merit of two galvanometers G_1 and G_2 of resistances 100 Ω and 50 Ω, are 10^{-8} A/div and 2×10^{-6} A/div respectively. Then voltage sensitivity of

a. G_1 is more than G_2
b. G_2 is more than G_1
c. Equal for G_1 and G_2
d. G_1 and G_2 cannot be compared

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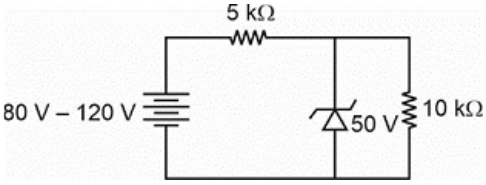
46. The I - V characteristics of a PN junction diode is shown below. The approximate dynamic resistance of the P-N junction when a forward bias of 6 V is applied



a. 1.5 Ω
b. 2.5 Ω
c. 25 Ω
d. 20 Ω

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47. For the circuit shown below, if I_z is current in Zener diode, then minimum value of I_z is



a. 5 mA
b. 6 mA
c. 1 mA
d. 2 mA

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48. To determine the refractive index of glass slab using a travelling microscope, minimum number of readings required are

a. Two
b. Four
c. Three
d. Five

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49. A multimeter is connected across the two terminals of a device under resistance mode. If pointer moves when voltage is applied one way and also when connection is reversed, then the device is

a. Resistor
b. Capacitor
c. LED
d. Diode

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50. The main scale of vernier calliper in mm has 19 division matched with 20 division of vernier scale. In measuring the diameter of cylinder by the instrument, the main scale reads with 32 division and 6th division of vernier scale coincide with main scale division. The diameter of cylinder is

a. 3.26 cm
b. 3.23 cm
c. 3.03 cm
d. 3.42 cm

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51. Appearance of violet colour on addition of sodium nitroprusside in the sodium fusion extract indicates the presence of

a. Chlorine

52. Prussian blue colour observed during detection of nitrogen is due to the formation of

a. $[Fe(NCS)]^{2+}$
b. $Fe_4[Fe(CN)_6]_3$

b. Bromine
c. Carbon
d. Sulphur

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c. $[Fe(CN)_6]^{3-}$
d. $[Fe(CN)_5NOS]^{4-}$

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53. Colour of $[Fe(SCN)]^{2+}$ is

a. Yellow
b. Blood red
c. Orange
d. Deep blue

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54. Layer test is done for detection of

a. N
b. S
c. C
d. Br

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55. Alcoholic compounds on reaction with ceric ammonium nitrate gives

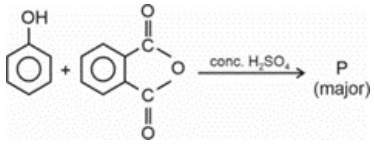
a. Yellow colour
b. Red colour
c. Green colour
d. Blue colour

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56. Which among the following will not give iodoform test?

a. CH_3CHO
b. CH_3COCH_3
c. CH_3CH_2OH
d. HCHO

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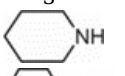
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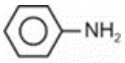
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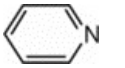
a. Pthalic acid
b. Phenolphthalein
c. Benzoic acid
d. Orthohydroxybenzoic acid

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58. Carbylamine test will be given by

a. CH_3NHCH_3
b. 

c. 

d. 

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59. Mohr's salt is

a. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
b. $Fe_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O$
c. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 24H_2O$
d. $FeSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

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60. Given below are two statements, one is labelled as assertion (A) other is labelled as reason (R)

Assertion (A): Preparation of acetanilide is done in presence of strong base.

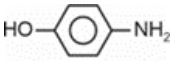
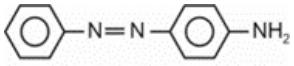
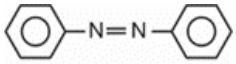
Reason (R): In the preparation of acetanilide bases stronger than aniline can remove HCl formed to shift equilibrium forward.

In the light of the above statements, choose the correct answer from the options given below.

a. Both A and R are true but R is not the correct explanation of A
b. A is true but R is false
c. A is false but R is true
d. Both A and R are true and R is the correct explanation of A

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61. Aniline yellow is

a. 
b. 
c. 
d. 

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62. Colour of phenolphthalein in acidic medium is

a. Pink
b. Orange
c. Blue
d. Colourless

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63. Mass of hydrated oxalic acid required to prepare 0.1 molar 100 ml solution will be

a. 12.6 g
b. 1.26 g
c. 10.8 g
d. 1.08 g

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64. The pair of compounds which can be distinguished by iodoform test is

a. Acetophenone and Acetone
b. Ethanol and Ethanal
c. Benzaldehyde and Isopropyl alcohol
d. Benzoic acid and benzyl alcohol

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65. Metal ion that exhibit characteristic light pink colour is

a. Cu^{2+}
b. Ni^{2+}
c. Fe^{2+}
d. Mn^{2+}

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66. Prussian blue colour is observed on addition of $K_4[Fe(CN)_6]$ in

a. $ZnCl_2$
b. $FeCl_3$
c. $CuCl_2$
d. $AlCl_3$

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67. Cation that gives brown residue on charcoal cavity test is

a. Pb^{2+}
b. Cd^{2+}
c. As^{3+}
d. Zn^{2+}

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68. Which among the following is not a group IV cation?

a. Pb^{2+}
b. Ni^{2+}
c. Zn^{2+}
d. Co^{2+}

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69. Cation that gives brown precipitate with Nessler's reagent is

a. Cu^{2+}
b. Zn^{2+}
c. NH_4^+
d. Mg^{2+}

70. Color of PbI_2 precipitate is

a. Black
b. Blue
c. White
d. Yellow

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71. Given below are two statements.

Statement I: $PbSO_4$ is highly soluble in water.

Statement II: $PbSO_4$ forms tetraacetoplumbate (II) with ammonium acetate solution.

In the light of above statements choose the **correct** option.

a. Both the statements I and II are correct
b. Both the statements I and II are incorrect
c. Statement I is incorrect and statement II is correct
d. Statement I is correct and statement II is incorrect

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72. Brown ring test is used to detect

a. S^{2-}
b. NO_3^-
c. SO_4^{2-}
d. Cl^-

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73. Sodium nitroprusside is

a. $Na_4[Fe(CN)_5NOS]$
b. $Na_4[Fe(CN)_6]$
c. $Na_2[Fe(CN)_5NO]$
d. $Na_3[Fe(CN)_4(NOS)_2]$

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74. Match groups in list-1 with their group reagent in list-2 and choose correct answer.

	List-1		**List-2**
(a)	Group-I	(i)	NH_4OH in presence of NH_4Cl
(b)	Group-II	(ii)	H_2S in presence of NH_4OH
(c)	Group-III	(iii)	Dilute HCl
(d)	Group-IV	(iv)	H_2S gas in presence of dil. HCl

a. (a)(iii), (b)(iv), (c)(i), (d)(ii)
b. (a)(i), (b)(ii), (c)(iii), (d)(iv)
c. (a)(iv), (b)(iii), (c)(ii), (d)(i)
d. (a)(ii), (b)(i), (c)(iv), (d)(iii)

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75. On addition of dil H_2SO_4 in a salt solution, a gas is evolved which turns lime water milky. Anion present in salt may be

a. NO_3^-
b. CO_3^{2-}
c. Cl^-
d. Br^-

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76. Which among the following sulphide has maximum solubility in water?

a. PbS
b. CuS
c. NiS
d. As_2S_3

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77. A gas that turns lead acetate paper black is

a. NO_2
b. HCl
c. CO_2
d. H_2S

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78. Anion that gives smell of rotten egg with warm dil H_2SO_4 is

a. SO_3^{2-}
b. NO_2^-
c. Cl^-
d. S^{2-}

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79. A salt on heating with conc. H_2SO_4 and MnO_2 liberates reddish brown fumes. The evolved brown fumes are of

a. N_2
b. I_2
c. Cl_2
d. Br_2

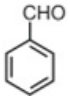
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80. Dimethyl glyoxime in basic medium is used to test

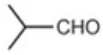
a. Co^{2+}
b. Ni^{2+}
c. Fe^{2+}
d. Zn^{2+}

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81. Which among the following compounds does not give positive Fehling test?

a. 

b. 

c. 

d. HCHO

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82. Propanal when treated with Schiff's reagent then the colour of the solution obtained is

a. Blue
b. Pink
c. Yellow
d. Green

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83. Addition of $AgNO_3$ in a salt solution gives white precipitate, solution may contain

a. Cl^-
b. Br^-
c. I^-
d. PO_4^{3-}

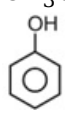
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84. Volume of 0.1 N HCl required for neutralisation of 10 mL 1 M NaOH solution will be

a. 1 mL
b. 10 mL
c. 100 mL
d. 1000 mL

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85. Which compound on reaction with $NaHCO_3$ gives CO_2 gas?

a. CH_3COOH
b. $CH_3 - O - CH_3$
c. CH_3CH_2OH
d. 

86. The colour of the solution obtained on addition of iodine to the starch solution is

a. Red
b. Green
c. Orange
d. Blue

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87. In the flame test, which of the given ions appear crimson red to the naked eye?

a. Cu^{2+}
b. Sr^{2+}
c. Ba^{2+}
d. Ca^{2+}

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88. Group reagent for Ba^{2+} cation is

a. $H_2S + NH_4OH$
b. $(NH_4)_2CO_3 + NH_4OH$
c. $NH_4OH + NH_4Cl$
d. HCl(aq)

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89. Consider the following statements about colloids

(a) Generally, size of colloidal particles ranges from 1–1000 nm

(b) Starch sol is an example of lyophilic sol.

(c) Stability of lyophobic sol is due to charge on colloidal particles.

The correct statements are

a. (a) and (b) only
b. (b) and (c) only
c. (a) and (c) only
d. (a), (b) and (c)

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90. Match list-1 with list-2 for given balanced chemical reaction and choose correct answer.

$$xMnO_4^- + yC_2O_4^{2-} + zH^+ \rightarrow xMn^{2+} + uCO_2 + vH_2O$$

List-1		List-2	
(i)	x	(a)	10
(ii)	y	(b)	2
(iii)	z	(c)	5
(iv)	u	(d)	16

a. (i)(a), (ii)(b), (iii)(c), (iv)(d)
b. (i)(d), (ii)(c), (iii)(b), (iv)(a)
c. (i)(b), (ii)(c), (iii)(a), (iv)(d)
d. (i)(b), (ii)(c), (iii)(d), (iv)(a)

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91. The element which does not give Lassaigne test is

a. B
b. N
c. P
d. I

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92. Consider the following statements

(a) AgI is insoluble in ammonia solution

(b) Na_2S on treatment with concentrated HNO_3 give SO_2 gas

(c) AgCl is soluble in excess of ammonia solution.

The correct statement are

a. (a) and (b) only
b. (a) and (c) only
c. (b) and (c) only
d. (a), (b) and (c)

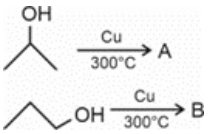
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93. Negatively charged sol among the following is

94. Reagent that can distinguish products A and B formed in given chemical reactions is

a. TiO_2 sol
b. Methylene blue sol
c. $Al_2O_3 \cdot xH_2O$ sol
d. As_2S_3 sol

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a. $HCl + ZnCl_2$
b. Tollens' reagent
c. 2, 4-DNP
d. NaH

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95. Chromyl chloride test can be given by

a. KNO_3
b. NaCl
c. Na_2SO_4
d. AgBr

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96. A salt gives violet vapours when treated with conc. H_2SO_4, it contains

a. I^-
b. Br^-
c. Cl^-
d. NO_3^-

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97. On addition of KCN in $CuSO_4$ solution, one of the products formed is

a. $[Cu(CN)_4]^{2-}$
b. $Cu(CN)_2$
c. $(CN)_2$
d. K_2SO_3

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98. On addition of β-naphthol to diazonium salt colour obtained is

a. Orange
b. Yellow
c. Scarlet red
d. Blue

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99. Working pH range of methyl orange indicator, is

a. 3.1 to 4.5
b. 4.2 to 6.2
c. 6.2 to 8.2
d. 8.2 to 10.2

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100. Na_2CrO_4 on reaction with NaOH and H_2O_2 forms compound [A]. [A] gets dissolved in amyl alcohol to give blue colour. The compound [A] is

a. Cr_2O_3
b. CrO_3
c. CrO_5
d. $Na_2Cr_2O_7$

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101. Which of the following statement is **not** related to sac-fungi?

a. They can be both multicellular and unicellular
b. They are saprophytic, decomposers, parasitic or coprophilous
c. Mycelium is aseptate and coenocytic
d. Sexual spores are produced endogenously

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102. The plasmodium differentiates under unfavourable conditions to form

a. Fruiting bodies
b. Vegetative structures
c. Zoospores
d. Conidia

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103. How many of the following is/are the function(s) of cell wall?

(a) Gives shape to the cells

(b) Protects from mechanical damage and infection

(c) Regulates the transport of the molecules across it

(d) Helps in cell to cell interaction

a. One
b. Two
c. Three
d. Four

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104. Which of the following cells has both 70S and 80S ribosomes?

a. *Lactobacillus*
b. *Rhizobium*
c. Mesophyll cell
d. *Nostoc*

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105. All of the following phenomenon may occur twice in meiosis, **except**

a. Spindle formation
b. Uncoiling of chromosomes
c. DNA replication
d. Centriole duplication

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106. In a chromosome, the primary constriction is present at the centre. Which statement is true about this chromosome?

a. It has two equal arms
b. It has two unequal arms
c. It has one small and one long arm
d. It has two small and two long arms

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107. Gymnosperms differ from ferns in having/being

a. Embryophytes
b. Vascular tissues
c. Ovule
d. Archegonia

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108. Select the **incorrect** statement w.r.t. bryophytes.

a. The plant body is attached to the substratum by unicellular or multicellular rhizoids
b. They possess stem-like and leaf-like structures
c. The sporophyte is always free-living and photosynthetic
d. Zygote does not undergo reduction division immediately

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109. Stem of maize is characterised by presence of

a. Open, conjoint vascular bundles
b. Starchy endodermis
c. Sclerenchymatous general cortex
d. Water containing cavities in the vascular bundles

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110. Epigynous flowers are found in

a. Rose
b. Brinjal
c. Ray florets of sunflower
d. Peach

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111. The cells of secondary medullary rays in dicot plants are formed by

a. Pith cells
b. Cambium
c. Hypodermal cells

112. The given question consists of two statements Assertion (A) and Reason (R). Answer the question by selecting appropriate option given below:

Assertion (A) : Nucleolus is the site of protein synthesis.

d. Endodermal cells

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Reason (R) : Larger and more numerous nucleoli are present in cells actively carrying out lipid synthesis.

a. (A) is true but (R) is false
b. Both (A) and (R) are true but (R) is not correct explanation of (A)
c. Both (A) and (R) are true and (R) is correct explanation of (A)
d. Both (A) and (R) are false

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113. Which type of flowers often have a single ovule in each ovary?

a. Animal-pollinated
b. Water-pollinated
c. Wind-pollinated
d. Insect-pollinated

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114. Read the following statements and choose the **correct** option.

Statement A: Dominance is an autonomous feature of a gene or the product that it has information for.

Statement B: Pea plant does not show incomplete dominance for any of the characters present in it.

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement A is correct
d. Only statement B is correct

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115. State true (T) or false (F) for the following statements and select the **correct** option.

(A) In *E.coli,* the process of replication requires a set of catalysts.
(B) Replication and transcription follow the rule of complementarity.
(C) DNA polymerase has free phosphate moiety at 3′ end of sugar.
(D) In prokaryotes, the DNA is scattered throughout the cell.

	A	**B**	**C**	**D**
(1)	T	T	T	T
(2)	T	F	T	F
(3)	T	T	T	F
(4)	T	T	F	F

a. (1)
b. (2)
c. (3)
d. (4)

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116. Which of the following is **not** found in RNA?

a. Pentose sugar
b. 5-methyl uracil
c. Phosphate group
d. N-glycosidic linkage

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117. Read the following statements and choose the **correct** option.

Statement A: The spread of living pteridophytes is limited and restricted to narrow geographical regions.

Statement B: Unicellular, free-living, mostly photosynthetic thalloid gametophyte of pteridophytes is called prothallus.

118. Who found that plots with more species showed less year to year variation in total biomass?

a. Humboldt
b. David Tilman
c. Paul Ehrlich
d. Robert Constanza

is called prothallus.

a. Only statement A is correct
b. Only statement B is correct
c. Both the statements are correct
d. Both the statements are incorrect

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Pdf By: @Loop2x

119. Which of the following is **incorrect** w.r.t. ecosystem?

a. Ecosystems are not exempt from the second law of thermodynamics
b. Herbaceous and woody plants are major producers in terrestrial ecosystem
c. Sun is the only source of energy for all ecosystems on Earth
d. The green plants in the ecosystem are called producers

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120. Find the **odd** one w.r.t. regulators.

a. Consume less amount of energy as compared to conformers
b. Capable of thermoregulation
c. Wide range of distribution
d. Maintain homeostasis

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121. Read the following statements and choose the **correct** option.

Assertion : Root cap protects the tender apex of the root.

Reason : Root hair increases the surface area for absorbtion of water and minerals from the soil.

a. Both Assertion & Reason are true and the reason is the correct explanation of the assertion
b. Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
c. Assertion is true statement but Reason is false
d. Both Assertion and Reason are false statements

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122. Which of the following is **not** a characteristic of arithmetic growth?

a. Constant growth rate
b. Linear growth curve
c. Presence of stationary phase
d. Commonly seen by root meristems

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123. Read the following statements and select the **correct** option.

Statement A: Pollen grains loose viability within 30 minutes of their release in all the members of Rosaceae, Leguminoseae and Solanaceae.

Statement B: Pollen grains can be stored in liquid nitrogen at –196° C for years.

a. Only statement A is correct
b. Only statement B is correct
c. Both statements A and B are correct
d. Both statements A and B are incorrect

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124. How many nuclei are involved in double fertilisation in an angiospermic plant?

a. Two
b. Three
c. Six
d. Five

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125. Which of the following traits in a pea plant can occur only in homozygous condition?

126. **Statement A:** A daughter will not normally be colour-blind, unless her mother is atleast a carrier and father is colour-blind.

a. Green pod colour
b. Yellow seed colour
c. Terminal flower position
d. Violet flower colour

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Statement B: A colour-blind man cannot pass the disorder to his sons but can pass on to grandsons.

Read the above statements and select the **correct** option.

a. Only statement A is correct
b. Only statement B is correct
c. Both statements A & B are incorrect
d. Both statements A & B are correct

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127. UTRs (untranslated regions) are some additional sequences present on mRNA which **do not** undergo translation. These sequences are present at

a. 5′end only (before start codon)
b. 5′end (after start codon) and 3′end (before stop codon)
c. 5′end (before start codon) and 3′end (after stop codon)
d. 3′end only (after start codon)

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128. Select the stage of sewage treatment which mainly involves microbes.

a. Primary treatment
b. Physical treatment
c. Secondary treatment
d. Tertiary treatment

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129. The mean annual precipitation will be highest in

a. Coniferous forest
b. Temperate forest
c. Grassland
d. Tropical rain forest

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130. Select the **correct** order of taxonomic category in ascending order.

a. Species → Genus → Class → Order → Family
b. Species → Genus → Family → Order → Class
c. Kingdom → Species → Class → Family → Order
d. Species → Family → Genus → Order → Class

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131. If proteins have to be respired, they would first be degraded by proteases into

a. PGAL
b. Amino acids
c. Acetyl CoA
d. Pyruvic acid

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132. Read the following statements and choose the **incorrect** one(s) w.r.t. chemiosmotic hypothesis.

(a) Breakdown of proton gradient leads to synthesis of ATP.

(b) CF_0 part of ATP synthase is embedded in thylakoid membrane and caries out facilitated diffusion of protons across the membrane.

(c) Breakdown of the gradient provides enough energy to cause a conformational change in the CF_0 particle of ATP synthase.

a. Only (c)
b. Only (b) and (c)
c. Only (a) and (b)
d. All (a), (b) and (c)

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133. The form of growth wherein new cells are always being added to the plant body by the meristem is called

a. Open
b. Closed
c. Determinate
d. Reversible

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134. Which of the following is **not** associated with photochemical phase of photosynthesis?

a. Light absorption
b. Water-splitting
c. Oxygen release
d. Formation of FADH and $NADH_2$

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135. Select the **correct** match w.r.t the IUCN red list (2004) documenting the number of extinct species.

a. 784 - Plants
b. 359 - Birds
c. 338 - Vertebrates
d. 87 - Invertebrates

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136. How many ATP molecules will be produced from the complete oxidation of one molecule of acetyl CoA?

a. 12
b. 14
c. 36
d. 38

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137. Cyclic photophosphorylation

a. Synthesises both ATP and NADPH
b. Synthesises ATP only
c. Synthesises NADPH only
d. Neither synthesise ATP nor NADPH

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138. The hormonal substance responsible for flowering migrates from (i) to (ii) for inducing flowering.

Select the **correct** option to fill in the blanks (i) and (ii).

a. (i) Apical buds, (ii) Axillary buds
b. (i) Roots, (ii) Shoot apices
c. (i) Leaves, (ii) Shoot apices
d. (i) Shoot apices, (ii) Axillary buds

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139. Structural genes in the *lac* operon of *E. coli* get switched on when

a. Repressor binds to the operator
b. Lactose binds to the RNA polymerase
c. Lactose binds to the repressor protein
d. Inducer binds to the operator

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140. In gymnosperms, the reduced male gametophyte which is confined to only limited number of cells is called

a. Strobili
b. Pollen grain
c. Ovule
d. Microsprophyll

Pdf By: @Loop2x

141. Companion cells are specialised

a. Parenchymatous cells
b. Sclerenchymatous cells
c. Collenchymatous cells
d. Enucleated cells

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142. Read the following statements and choose the **correct** option.
Statement A: The structure that helps in climbing in cucumber develops from apical bud.
Statement B: Flattened, chlorophyllous stems are present in *Pistia*

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement A is correct
d. Only statement B is correct

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143. Usage of terms growth and reproduction is not clear in which of the following organisms?

a. *Planaria*
b. Filamentous algae
c. Bryophytes
d. Unicellular algae

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144. Read the following statements and choose the **correct** option.
Statement A: Basidiospores are the exogenously produced sexual spores.
Statement B: Basidiospores are produced on sex organs.

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement A is correct
d. Only statement B is correct

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145. Match the following columns and choose the **correct** option.

	Column I		**Column II**
a.	Fluid-mosaic model	(i)	Golgi apparatus
b.	Calcium pectate	(ii)	Plasma membrane
c.	Divide by fission	(iii)	Middle lamella
d.	Formation of glycoproteins and glycolipids	(iv)	Mitochondria

a. a(i), b(iv), c(ii), d(iii)
b. a(iii), b(i), c(ii), d(iv)
c. a(ii), b(iii), c(iv), d(i)
d. a(i), b(ii), c(iii), d(iv)

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146. The organelle which forms the spindle fibers duplicates during

a. G phase
b. S phase
c. G_0 phase
d. G_2 phase

Pdf By: @Loop2x

147. *Trichoderma* species are free-living __A__ that are very common in __B__ ecosystems.

a. Fungi, root
b. Bacteria, root
c. Fungi, shoot
d. Bacteria, shoot

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148. α -Thalassemia is

a. Caused due to abnormality in chromosome 11
b. Commonly observed in females only
c. Controlled by two closely linked genes on chromosome 16
d. Controlled by HBA1 gene and HBB gene

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149. In both *Yucca* and *Amorphophallus*,

a. Abiotic agents perform pollination
b. Nectar robbers perform pollination
c. Their flowers provide safe place to the pollinators to lay eggs
d. No pollinating agent is required

150. The rate of formation of new organic matter by consumers is called

a. Secondary productivity
b. Primary productivity
c. Net primary productivity
d. Gross primary productivity

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Pdf By: @Romlex69

151. The common feature between *Branchiostoma*, *Scoliodon* and *Pristis* is

a. Persistent notochord throughout the life
b. Presence of dorsal heart
c. Absence of pharynx perforated by gill slits
d. Absence of scales and paired fins

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152. Choose the option that includes the organisms that possess pseudocoelom as well as well-developed muscular pharynx.

a. Earthworm, Tapeworm, Silkworm
b. Tapeworm, Roundworm, Hookworm
c. Hookworm, Filarial worm, Roundworm
d. Silkworm, Hookworm, Tapeworm

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153. The type of epithelium present in the inner lining of ducts of salivary glands is also found in the

a. Blood vessels
b. Pharynx
c. Small intestine
d. PCT of nephron

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154. The total number of fatty acids in a lecithin is equal to the total number of

a. Nitrogen atoms in uracil
b. Hydroxyl groups in glycerol
c. Hydrogen atoms in glycine
d. Hydroxyl groups in cholesterol

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155. The maximum volume of air a person can breathe in after a forced expiration is 'X'. 'X' includes all of the following, **except**

a. TV
b. ERV
c. RV
d. IRV

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156. In a standard ECG, QRS complex represents an event in which there is

a. Production of second heart sound characterised by low pitch
b. Rise in intraventricular pressure
c. Closure of semilunar valve
d. Flow of blood into the atria from ventricles

Pdf By: @Loop2x

157. Select the **incorrect** match w.r.t. excretory structure

a. Prawns – Green glands
b. *Pheretima* – Protonephridia
c. *Periplaneta* – Malpighian tubules
d. *Planaria* – Flame cells

Pdf By: @Romlex69

158. The type of movement that assists in locomotion of *Euglena* also assists in

a. Streaming of protoplasm in *Amoeba*
b. Passage of ova through the oviduct
c. Maintenance of water current in the canal system of *Sycon*
d. Capturing of prey by *Hydra*

Pdf By: @Loop2x

159. The hormone whose hyposecretion is associated with diabetes mellitus is involved in all of the following, **except**

a. Promoting both oxidation of glucose and conversion of glucose into glycogen
b. Inhibiting synthesis of fats from glucose by adipose tissue
c. Promoting uptake of amino acids by liver and muscle cells
d. Stimulating transport of glucose from blood to

160. Complete the analogy by selecting the correct option w.r.t. human females.

Female accessory duct : Uterus :: Female external genitalia : _____

a. Cervix
b. Vagina
c. Oviduct
d. Clitoris

muscle

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Pdf By: @Loop2x

161. Choose the STI among the following which can be cured if detected early and treated properly

a. Hepatitis-B
b. Genital herpes
c. Genital warts
d. HIV infections

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162. The scientist who dismissed the spontaneous generation theory showed that

a. In pre-sterilised flasks, life did not come from killed yeast
b. First form of life come from pre-existing non-living organic molecules
c. Unit of life called spores were transferred to different planets including Earth
d. The first cellular form of life did not possibly originate till about 2 mya

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163. The drug 'X' is a very effective sedative and painkiller . 'X' belongs to the category of drugs whose receptors are present in
(a) CNS
(b) Stomach
(c) Small intestine
Choose the **correct** option.

a. a and c only
b. b and c only
c. a, b and c
d. b only

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164. Assume a circular DNA 'P' of 15 kb size possess three recognition sites for *Sal* I at equal distance and if 'P' is completely digested using *Sal I*, then how many bands will be visible on separation of DNA fragments using agarose gel electrophoresis?

a. 1
b. 3
c. 4
d. 0

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165. The first transgenic cow, Rosie, produced human protein-enriched milk in the year 'Q'. Choose the event that also occurred in 'Q'.

a. Production of recombinant form of human insulin
b. First gene therapy to a 4-year old girl with ADA deficiency
c. An American company got patent rights on Basmati rice *via* US patent and trademark office
d. Isolation of enzymes responsible for restricting the growth of bacteriophage in *E.coli*

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166. Read the given statements and select the correct option.

Statement A: In the cloning vector pBR322, restriction sites for *EcoR* I, *Sal* I, *Pvu* ll and *Hind* III are not present within any antibiotic- resistance genes.

Statement B: If a foreign DNA is inserted at *BamH* I site in the vector pBR322, then the recombinant plasmids will lose ampicillin resistance.

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement A is correct
d. Only statement B is correct

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167. Male frogs can be distinguished from the female frogs by the presence of

a. Five digits in hindlimbs
b. Sound-producing vocal sacs
c. Webbed feet for swimming
d. Copulatory pad on the second digit of the forelimb

168. Signals from the pneumotaxic centre result in shallow and fast breathing. This centre is present in which region of hindbrain?

a. Pons
b. Cerebellum
c. Medulla oblongata
d. Hippocampus

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169. Consider the given statements and select the correct option w.r.t. neural system.

Statement A: Myelinated nerve fibres are enclosed by Schwann cells and are found in spinal and cranial nerves.

Statement B: Certain granular bodies called Nissl's granules are present in dendrites, cyton as well as in axon.

a. Both the statements are correct
b. Both the statements are incorrect
c. Only statement (A) is correct
d. Only statement (B) is correct

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170. Choose the hormone that peaks twice in a human menstrual cycle

a. Progesterone
b. LH
c. FSH
d. Estrogen

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171. A person 'X' went to the doctor for vasectomy. After the procedure, which of the following could be observed in 'X'?

a. Reduction in libido
b. Absence of sperm in semen
c. Inhibition of spermatogenesis
d. No change in the ejaculate

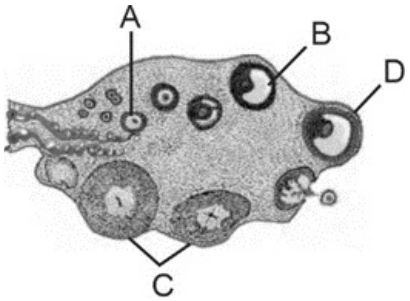
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172. Choose the odd one w.r.t. animals in which notochord is present only in larval tail

a. *Ascidia*
b. *Salpa*
c. *Doliolum*
d. *Carcharodon*

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173. Consider the given labelled diagrammatic sectional view of ovary. Select the correct option w.r.t. it



a. At puberty only 60,000 – 80,000 'A' are present in both ovaries
b. 'B is characterised by theca externa and theca interna as well as oogonium
c. 'C' is responsible for secreting progesterone as well as estrogen
d. 'D' possesses primary oocyte surrounded by zona pellucida

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174. *Aptenodytes* belong to a group of animals having

a. Mammary glands and 2 pairs of limbs
b. Epidermal scales with 3-chambered heart
c. Mixing of oxygenated and deoxygenated blood in ventricle
d. Oviparity and air sacs to supplement pulmonary respiration

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175. Hardy-Weinberg equilibrium remains unaffected

176. Which hormone is characterised by positive

by presence of which of the following condition?

a. Genetic drift
b. Mutation
c. Random mating
d. Recombination

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feedback regulation that drives the birth process to completion?

a. Estrogen
b. Progesterone
c. Oxytocin
d. Inhibin

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177. Choose the **incorrect** pair.

(1)	Epithelial tissue	–	Forms microvilli in alimentary canal
(2)	Adipose tissue	–	Functions as shock-absorber
(3)	Cartilage	–	Has hard and non-pliable matrix
(4)	Bone	–	Contains osteoblasts

a. (1)
b. (2)
c. (3)
d. (4)

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178. How many of the following are chemically proteins?

Collagen, Trypsin,
Insulin, Antibody, Receptor,
GLUT-4

Select the correct option.

a. Four
b. Six
c. Five
d. One

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179. If a molecule of carbon dioxide released into the blood in your left foot travels up to your lungs, it must pass through all of the following structures, **except**

a. Right atrium
b. Pulmonary vein
c. Inferior vena cava
d. Pulmonary artery

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180. Complete the following reaction

$\text{Prothrombin} \xrightarrow{A} \text{Thrombin}$

$B \xrightarrow{\text{Thrombin}} \text{Fibrin}$

Choose the **correct** option w.r.t. A and B.

a. A – Calcium B – Thrombokinase
b. A – Fibrinogen B – Thromboplastin
c. A – Thrombokinase B – Fibrinogen
d. A – Fibrinogen B – Thrombokinase

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181. Consider the following four statements A, B, C and D and select the **correct** option which states whether they are true (T) or false (F).

A. An excessive loss of fluid from body stimulates the osmoreceptors which stimulate hypothalamus to release antidiuretic hormone (ADH) from the neurohypophysis

B. Rennin converts angiotensinogen to angiotensin I

C. Urea is reabsorbed in the entire loop of Henle.

D. Water reabsorption is minimum in proximal convoluted tubule.

	A	**B**	**C**	**D**
(1)	T	F	F	F
(2)	T	T	F	F
(3)	F	F	F	T
(4)	T	F	T	F

182. The thick filaments of A band are held together by a thin fibrous membrane called as

a. ‘I’ band
b. ‘M’ line
c. ‘Z’ line
d. ‘H’ zone

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(4) I r I r

a. (1)
b. (2)
c. (3)
d. (4)

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183. Repolarisation of a nerve fibre is characterised by the

a. Rapid influx of K^+ ions
b. Rapid efflux of K^+ ions
c. Rapid efflux of Ca^{+2} ions
d. Rapid influx of Na^+ ions

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184. Select the **odd one** w.r.t hormones secreted by pars distalis portion of pituitary gland.

a. LH
b. ADH
c. FSH
d. TSH

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185. A female who is on minipills, stops taking them after 21st day of her normal menstrual cycle. Which of the following will be the effect?

a. It results in Pregnancy during this menstrual cycle
b. Fertility is lost
c. It leads to menstrual bleeding
d. Menstruation stops

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186. The second meiotic division of the mammalian ovum is completed

a. In the Graafian follicle following the first maturation division
b. Shortly after ovulation before the ovum makes entry into the fallopian tube
c. When the ovum is penetrated by a sperm
d. Until the nucleus of the sperm has fused with that of the ovum

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187. What is the action of progestogens present in the oral contraceptive pills? Choose the **correct** option.

(a) Inhibits implantation

(b) Inhibits ovulation

(c) Alter the quality of cervical mucus

(d) Prevent / retard the entry of sperm

a. (a) and (b) only
b. (a) and (c) only
c. (b), (c) and (d) only
d. (a), (b), (c) and (d)

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188. Consider the below given lymphoid organs
a. Bone marrow
b. Spleen
c. Thymus
d. MALT
e. Lymph nodes
Which of them are primary lymphoid organs?

a. a, b and c
b. a and c
c. a, d and e
d. b, d and e

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189. Which of the following is associated with gel electrophoresis?

a. Spooling
b. Elution
c. Transformation
d. Separation and isolation of recombinant product

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190. Which of the following enzymes is used for treating a bacterial cell for removal of its cell wall?

a. Lysozyme
b. Cellulase
c. Chitinase
d. Amylase

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191. Genetically modified plants have been useful in all of the following ways, **except**

a. Increased efficiency of mineral usage
b. Reduced post-harvest losses
c. Crops more tolerant to abiotic stresses
d. Increased reliance on chemical pesticides

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192. Number of enzymes have non-protein constituents called co-factor. In these instances, the protein portion of the enzyme is called

a. Apoenzyme
b. Co-enzyme
c. Holoenzyme
d. Prosthetic group

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193. All of the following structures are involved in systemic circulation, **except**

a. Dorsal aorta, capillary
b. Capillary, vena cava
c. Right atrium, left ventricle
d. Left atrium, right ventricle

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194. On an average, _A_ mL of blood is filtered by the kidneys per minute which constitutes roughly _B_ of the blood pumped out by each ventricle of the heart in a minute.

Choose the option which **correctly** fills the blanks A and B respectively.

a. 125, 1/2
b. 1100 – 1200, $1/5^{th}$
c. 700 – 800, $1/4^{th}$
d. 1100 – 1200, $3/4^{th}$

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195. Spread of action potential causes the release of which ions into the sarcoplasm?

a. Mg^{2+}
b. Na^{+}
c. Ca^{2+}
d. K^{+}

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196. Which of the following is in contact with the cranial meninx, pia mater?

a. White matter of brain only
b. Gray matter of spinal cord only
c. Gray matter of brain
d. White matter of both spinal cord and brain

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197. Among the following hormones that does not show tropic effect is

a. FSH, LH
b. TSH, ACTH
c. Oxytocin, vasopressin
d. LH, TSH

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198. The frogs differ from humans by possessing

a. Dicondylic skull
b. Urea as nitrogenous waste
c. Nucleated RBCs
d. Myogenic heart

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199. Which of the following is the second largest phylum in the animal kingdom?

a. Arthropoda
b. Echinodermata
c. Mollusca
d. Annelida

200. Read the following statements and choose the **correct** answer.
A. Transgenic mice are being used for testing safety of polio vaccine.
B. *Pseudomonas putida* is genetically engineered microorganism utilised for cleaning oil spills.

a. Both statements are correct

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a. Both statements are correct
b. Both statements are incorrect
c. Only statement A is correct
d. Only statement B is correct

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